



Mangroves Plastic Pollution

Study finds mangrove ecosystems are at risk of being polluted from plastic waste carried from rivers to the sea. <http://digit.in/feb21-03>



Air Pollution Linked To Sight Loss

Long-term study links air pollution to increased risk of irreversible age-related macular degeneration (AMD). <http://digit.in/feb21-04>

with increased climate change. The changes introduced by the nutrient pollution is ultimately expected to reduce biodiversity in the region, and increase the chances of wildfires. The research does show that the different grasslands around the world are impacted to different degrees, and domestic grazing animals, to a degree can reduce the damage caused. The researchers hope to investigate the specific ways in which domestic grazers can be used to counteract the effects of excessive nutrient pollution in the grasslands.

BLEACHED CORALS

Coral reefs are critical components of the aquatic system, and environmentalists working with coral reefs have to constantly monitor for the large range of environmental stressors that can negatively impact the coral. The corals are also particularly sensitive to changes in the environment. A new study showed that the survival and development of coral larvae were negatively affected by elevated levels of nutrients, and modest levels of rise in seawater temperatures, in the first few days of their life cycles. While we know of coral mostly by the hard rocky exoskeletons they leave behind, in the larval form the corals are very soft and gelatinous. The researchers subjected coral larvae to a range of nutrient conditions, as well as temperature changes. Elevated levels of nitrogen had the strongest impact on the coral, with survivorship dropping drastically. Adding phosphorus into the mix reduced the negative effect of the nitrogen alone. Subsequently heating the water did not reduce the survivorship, but affected the creatures in other ways, mostly by reducing the size of the larva. The study clearly showed that an excessive amount of nutrients can harm the coral, but more importantly also showed that an imbalance in the nutrients, combined with the temperature of the water can also negatively impact the development of the coral in the early stages of its life cycle.

Chris Wall, study co-author and postdoctoral researcher at SOEST's Pacific Biosciences Research Center said, "The study also revealed that the way corals reproduce and other biological differences among larvae influence how they respond to environmental change. The impacts of warming and elevated nutrients were greatest on small larvae that develop outside the parents, called broadcast spawners, and dependent on whether the larvae had already formed a relationship with their symbiotic algae. And interestingly, those larger larvae that develop inside their

derived from this carbon from plant matter. There are two main sources of the forest carbon in streams. One is through algal blooms, the other being the bits and pieces of plants that fall into the rivers from the forests around them, providing a steady source of food all around the year. While it was previously known that the addition of phosphorus and nitrogen to river streams can encourage excessive algal growth, causing a nuisance to other life forms sharing the stream, the new research shows that they also accelerate the breakdown of leaves and twigs in the stream, retaining the

natural forest carbon for a reduced period of time. When all parts of the forest and the stream are in natural balance, microbes, insects and other arthropods break down the leaves and twigs in the stream, and nitrogen and phosphorus play a crucial role in this process. It is only when these nutrients are introduced in excessive quantities, that they become pollutants

start becoming a hazard for all the living creatures in the environment. If you have been doing some mental math trying to come to a conclusion, not all carbon in the ecosystem is a zero sum game. This is because each component in the process that introduces the carbon to the system has a role to play in the ecosystem, the forests, the microbes, the arthropods and the algae. The excessive algal growth can smother other life forms on the streams. Ironically, it is the leaves and the twigs in the stream that help capture the nitrogen and phosphorus, but with too much of it, all the carbon and the excessive nutrients head further downstream, where they may reach coastal areas, estuaries and lakes. The researchers hope that their findings will guide policy decisions to reduce nutrient pollution of streams. **4**



Credit: Raphael Ritson-Williams

A settling coral on top and a swimming coral on the bottom. The tiny brown dots on the coral are symbiotic algae.

parents, called brooders, were quite resistant to warming and nutrient spikes." The researchers have shown how an excess of nutrients can negatively affect coral larvae, and are next hoping to study the impact throughout the life cycle of the coral, to get a complete picture.

DAMAGING STREAMS

A team of researchers led by University of Georgia ecologists, have uncovered the various ways in which nutrient pollution is negatively impacting rivers. No one had suspected some of the consequences. The residence time of the carbon from leaves, plants, twigs and other plant detritus is chopped by half even when moderate amounts of phosphorus and nitrogen are introduced to a water body. Much of the energy that drives food webs in rivers and streams, is